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M1 MACRO

Group A

Session 7

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

With N agents, in a ^{static} edowment econ,
A Pareto . optimal allocation $\{C_i\}_{i=1}^N$ is
solving the problem

$$\left[\begin{array}{l} \max \sum \omega_j u(C_j) \\ \text{st.} \sum_{j=1}^N C_j = \sum_{j=1}^N y_j \end{array} \right.$$

for given set of weights ω_j , $\omega_j > 0$
 $\sum \omega_j = 1$

If $N = 1$, Pareto optimum:

$$\begin{cases} \max u(c_1) \\ \text{s.t. } c_1 \leq y_1 \end{cases}$$

$\hookrightarrow$ only 1 solution $c_1 = y_1$

Optimal allocations

$$\text{Max}_{\{C_1, C_2, K\}} \log C_1 + \beta \log C_2$$

$$\text{s.t. } \begin{cases} C_1 + K = Y_1 \Rightarrow C_1 = Y_1 - K \\ C_2 = K^\alpha + Y_2 \end{cases}$$

$$\hookrightarrow \text{Max}_K \log(Y_1 - K) + \beta \log(K^\alpha + Y_2)$$

$$\text{FOC } / K : -\frac{1}{Y_1 - K} + \beta \frac{\alpha K^{\alpha-1}}{K^\alpha + Y_2} = 0$$

$$\Rightarrow \alpha \beta (Y_1 - K) K^{\alpha-1} = K^\alpha + Y_2$$

Assume: $y_1 = \bar{y}$, $y_2 = 0$

$$\alpha \beta y_1 K^{\alpha-1} - \alpha \beta K^{\alpha} = K^{\alpha} + y_2$$

$$\alpha \beta y_1 - \alpha \beta K = K + y_2 K^{1-\alpha}$$

$$K(1 + \alpha \beta) = \overline{\alpha \beta y}$$

$$K = \frac{\alpha \beta}{1 + \alpha \beta} \bar{y}$$

$$= \omega \bar{y}$$

Similar to Comput. Eq.

Note How to derive the interest rate
in the production economy:

$$\left\{ \begin{array}{l} \text{Max } \log c_1 + \beta \log c_2 \end{array} \right.$$

$$\text{s.t. } c_1 + X + S = Y_1$$

$$c_2 = Y_2 + X^\alpha + (1+r)S$$

→ for

equilibrium: $S=0$

$$\begin{aligned} c_1 + X &= Y_1 \\ c_2 &= Y_2 + X^\alpha \end{aligned} \quad + \quad \frac{1}{1+r} = \beta \frac{c_1}{c_2}$$